

Blockchain Experiment 11

AIM: To implement a Hyperledger Fabric network, deploy chaincode, and perform asset creation, querying, and transfer operations.

PROCEDURE

Step 1: Update the system

Go to <https://shell.cloud.google.com> and run the following commands:

```
sudo apt update
```

```
sudo apt upgrade -y
```

```
g2022_anuprita_mhapankar@cloudshell:~$ sudo apt update
Get:1 https://cli.github.com/packages stable InRelease [3,917 B]
Hit:2 https://download.docker.com/linux/ubuntu noble InRelease
Hit:3 https://apt.postgresql.org/pub/repos/apt noble-pgdg InRelease
Hit:4 https://packages.cloud.google.com/apt gcsfuse-noble InRelease
Hit:5 https://packages.cloud.google.com/apt cloud-sdk InRelease
Hit:6 https://repo.mongodb.org/apt/ubuntu noble/mongodb-org/8.0 InRelease
Hit:7 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:8 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:9 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:10 https://ppa.launchpadcontent.net/dotnet/backports/ubuntu noble InRelease
Hit:11 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Fetched 3,917 B in 6s (639 B/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
55 packages can be upgraded. Run 'apt list --upgradable' to see them.
```

```
g2022_anuprita_mhapankar@cloudshell:~$ sudo apt upgrade -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
The following upgrades have been deferred due to phasing:
 libpam-systemd libsystemd-shared libsystemd0 libudev1 qemu-user qemu-user-static systemd systemd-dev systemd-sysv udev
The following packages will be upgraded:
 google-cloud-cli-app-engine-go google-cloud-cli-app-engine-java google-cloud-cli-app-engine-python google-cloud-cli-app-engine-python-extras
 google-cloud-cli-bigtable-emulator google-cloud-cli-cbt google-cloud-cli-cloud-run-proxy google-cloud-cli-datastore-emulator google-cloud-cli-gke-gcloud-auth-plugin
 google-cloud-cli-istioctl google-cloud-cli-kpt google-cloud-cli-local-extract google-cloud-cli-minikube google-cloud-cli-nomes google-cloud-cli-package-go-module
 google-cloud-cli-pubsub-emulator google-cloud-cli-stafford google-cloud-cli-spanner-cli kubect1 libarchive13t64 libgdk-pixbuf-2.0-0 libgdk-pixbuf2.0-common libpq-dev
 libpq5 libruby3.2 linux-libc-dev php8.3-bcmath php8.3-cgi php8.3-cli php8.3-common php8.3-dev php8.3-mbstring php8.3-mysql php8.3-openssl php8.3-readline php8.3-xml
 ruby3.2 ruby3.2-dev
38 upgraded, 0 newly installed, 0 to remove and 10 not upgraded.
10 not fully installed or removed.
Need to get 603 MB of archives.
After this operation, 3,519 kB disk space will be freed.
Get:1 https://apt.postgresql.org/pub/repos/apt noble-pgdg/main amd64 libpq-dev amd64 18.3-1.pgdg24.04+1 [167 kB]
Get:2 https://apt.postgresql.org/pub/repos/apt noble-pgdg/main amd64 libpq5 amd64 18.3-1.pgdg24.04+1 [255 kB]
Get:3 https://packages.cloud.google.com/apt cloud-sdk/main all google-cloud-cli-app-engine-python all 564.0.0-0 [4,032 kB]
Get:4 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libarchive13t64 amd64 3.7.2-2ubuntu0.6 [382 kB]
Get:5 https://packages.cloud.google.com/apt cloud-sdk/main amd64 google-cloud-cli-app-engine-go amd64 564.0.0-0 [3,007 kB]
Get:6 https://packages.cloud.google.com/apt cloud-sdk/main all google-cloud-cli-app-engine-java all 564.0.0-0 [160 MB]
Get:7 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libgdk-pixbuf2.0-common all 2.42.10+dfsg-3ubuntu3.3 [8,302 B]
Get:8 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libgdk-pixbuf-2.0-0 amd64 2.42.10+dfsg-3ubuntu3.3 [147 kB]
Get:9 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 ruby3.2-dev amd64 3.2.3-1ubuntu0.24.04.7 [399 kB]
Get:10 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libruby3.2 amd64 3.2.3-1ubuntu0.24.04.7 [5,341 kB]
Get:11 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 linux-libc-dev amd64 6.8.0-107.107 [2,096 kB]
Get:12 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 php8.3-xml amd64 8.3.6-0ubuntu0.24.04.8 [126 kB]
Get:13 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 php8.3-readline amd64 8.3.6-0ubuntu0.24.04.8 [13.5 kB]
Get:14 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 php8.3-openssl amd64 8.3.6-0ubuntu0.24.04.8 [371 kB]
```

Step 2: Install Required Dependencies

```
g2022_anuprita_mhapankar@cloudshell:~$ sudo apt install -y jq curl git
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
jq is already the newest version (1.7.1-3ubuntu0.24.04.1).
curl is already the newest version (8.5.0-2ubuntu10.8).
git is already the newest version (1:2.43.0-1ubuntu7.3).
0 upgraded, 0 newly installed, 0 to remove and 10 not upgraded.
g2022_anuprita_mhapankar@cloudshell:~$
```

Then verify everything is present:

```
node -v
```

```
npm -v
```

```
git --version
```

```
docker --version
```

```
docker ps
```

```
g2022_anuprita_mhapankar@cloudshell:~$ node -v
v24.14.1
g2022_anuprita_mhapankar@cloudshell:~$ npm -v
11.12.1
g2022_anuprita_mhapankar@cloudshell:~$ git --version
git version 2.43.0
g2022_anuprita_mhapankar@cloudshell:~$ docker --version
Docker version 29.4.0, build 9d7ad9f
g2022_anuprita_mhapankar@cloudshell:~$ docker ps
CONTAINER ID   IMAGE      COMMAND                  CREATED        STATUS        PORTS        NAMES
g2022_anuprita_mhapankar@cloudshell:~$
```

Step 3: Fix Docker Compose

```
DOCKER_CONFIG=${DOCKER_CONFIG:-$HOME/.docker}
```

```
mkdir -p $DOCKER_CONFIG/cli-plugins
```

```
curl -SL
```

```
https://github.com/docker/compose/releases/download/v2.24.5/docker-compose
-linux-x86_64 \
```

```
  -o $DOCKER_CONFIG/cli-plugins/docker-compose
```

```
chmod +x $DOCKER_CONFIG/cli-plugins/docker-compose
```

```
docker compose version
```

```
sudo mv /usr/bin/docker-compose /usr/bin/docker-compose-old 2>/dev/null ||
true
```

```
sudo tee /usr/local/bin/docker-compose > /dev/null <<'EOF'
```

```
#!/bin/bash
```

```
exec docker compose "$@"
```

```
EOF
```

```
sudo chmod +x /usr/local/bin/docker-compose
```

```
docker compose version
```

docker-compose version

```
g2022_anuprita_mhapankar@cloudshell:~$ DOCKER_CONFIG=${DOCKER_CONFIG:-$HOME/.docker}
mkdir -p $DOCKER_CONFIG/cli-plugins
curl -SL https://github.com/docker/compose/releases/download/v2.24.5/docker-compose-linux-x86_64 \
-o $DOCKER_CONFIG/cli-plugins/docker-compose
chmod +x $DOCKER_CONFIG/cli-plugins/docker-compose
docker compose version
% Total      % Received % Xferd Average Speed   Time    Time     Time  Current
             Dload  Upload   Total   Spent    Left     Speed
  0     0    0     0     0     0      0  0  --:--:-- --:--:-- --:--:--    0
100 58.5M 100 58.5M    0     0 79.4M    0  --:--:-- --:--:-- --:--:-- 79.4M
Docker Compose version v2.24.5
g2022_anuprita_mhapankar@cloudshell:~$ sudo mv /usr/bin/docker-compose /usr/bin/docker-compose-old 2>/dev/null || true
g2022_anuprita_mhapankar@cloudshell:~$ sudo tee /usr/local/bin/docker-compose > /dev/null <<'EOF'
#!/bin/bash
exec docker compose "$@"
EOF
sudo chmod +x /usr/local/bin/docker-compose
g2022_anuprita_mhapankar@cloudshell:~$ docker compose version
docker-compose version
Docker Compose version v2.24.5
Docker Compose version v2.24.5
g2022_anuprita_mhapankar@cloudshell:~$
```

Step 4: Clone Fabric Samples

```
cd $HOME
git clone https://github.com/hyperledger/fabric-samples
cd fabric-samples
```

```
g2022_anuprita_mhapankar@cloudshell:~$ cd $HOME
g2022_anuprita_mhapankar@cloudshell:~$ git clone https://github.com/hyperledger/fabric-samples
Cloning into 'fabric-samples'...
remote: Enumerating objects: 15618, done.
remote: Counting objects: 100% (238/238), done.
remote: Compressing objects: 100% (129/129), done.
remote: Total 15618 (delta 161), reused 109 (delta 109), pack-reused 15380 (from 3)
Receiving objects: 100% (15618/15618), 24.07 MiB | 17.13 MiB/s, done.
Resolving deltas: 100% (8809/8809), done.
g2022_anuprita_mhapankar@cloudshell:~$ cd fabric-samples
g2022_anuprita_mhapankar@cloudshell:~/fabric-samples$
```

Step 5: Install Fabric Binaries and Docker Images

```
curl -sSLO
https://raw.githubusercontent.com/hyperledger/fabric/main/scripts/install-
fabric.sh
chmod +x install-fabric.sh
./install-fabric.sh docker samples binary
```

```
g2022 anuprita_mhapankar@cloudshell:~/fabric-samples$ curl -sLO https://raw.githubusercontent.com/hyperledger/fabric/main/scripts/install-fabric.sh
g2022 anuprita_mhapankar@cloudshell:~/fabric-samples$ chmod +x install-fabric.sh
./install-fabric.sh docker samples binary
```

Clone hyperledger/fabric-samples repo

```
==> Already in fabric-samples repo
fabric-samples v2.5.15 does not exist, defaulting to main. fabric-samples main branch is intended to work with recent versions of fabric.
```

Pull Hyperledger Fabric binaries

```
====> Downloading version 2.5.15 platform specific fabric binaries
====> Downloading: https://github.com/hyperledger/fabric/releases/download/v2.5.15/hyperledger-fabric-linux-amd64-2.5.15.tar.gz
====> Will unpack to: /home/g2022_anuprita_mhapankar/fabric-samples
% Total    % Received % Xferd Average Speed   Time    Time     Time  Current
           Dload  Upload   Total   Spent    Left     Speed
  0     0    0     0     0     0      0      0  --:--:-- --:--:-- --:--:--    0
100 120M 100 120M    0     0 22.5M    0  0:00:05 0:00:05 --:--:-- 24.4M
==> Done.
====> Downloading version 1.5.17 platform specific fabric-ca-client binary
====> Downloading: https://github.com/hyperledger/fabric-ca/releases/download/v1.5.17/hyperledger-fabric-ca-linux-amd64-1.5.17.tar.gz
====> Will unpack to: /home/g2022_anuprita_mhapankar/fabric-samples
% Total    % Received % Xferd Average Speed   Time    Time     Time  Current
           Dload  Upload   Total   Spent    Left     Speed
  0     0    0     0     0     0      0      0  --:--:-- --:--:-- --:--:--    0
100 31.6M 100 31.6M    0     0 15.9M    0  0:00:01 0:00:01 --:--:-- 31.6M
==> Done.
```

Pull Hyperledger Fabric docker images

```
FABRIC_IMAGES: peer orderer ccenv baseos
====> Pulling fabric Images
====> ghcr.io/hyperledger/fabric-peer:2.5.15
```

```
====> ghcr.io/hyperledger/fabric-ca:1.5.17
1.5.17: Pulling from hyperledger/fabric-ca
18afb8c8d2a4: Pull complete
6f4ebca3e823: Pull complete
b9649d3f054a: Pull complete
0b966e606d9a: Pull complete
c36bacd85367: Download complete
Digest: sha256:453a0a727e82c4960b797e379adc231e853ee23ae7526db53afef77b5074117e
Status: Downloaded newer image for ghcr.io/hyperledger/fabric-ca:1.5.17
ghcr.io/hyperledger/fabric-ca:1.5.17
====> List out hyperledger images
WARNING: This output is designed for human readability. For machine-readable output, please use --format.
ghcr.io/hyperledger/fabric-baseos:2.5.15    654bd4554bf9    250MB    80.4MB
ghcr.io/hyperledger/fabric-ca:1.5.17       453a0a727e82    381MB    123MB
ghcr.io/hyperledger/fabric-ccenv:2.5.15    86dded7eda92    1GB      255MB
ghcr.io/hyperledger/fabric-orderer:2.5.15  9972c209be47    178MB    49.5MB
ghcr.io/hyperledger/fabric-peer:2.5.15     3e25adc31a75    230MB    66.2MB
hyperledger/fabric-baseos:2.5              654bd4554bf9    250MB    80.4MB
hyperledger/fabric-baseos:2.5.15           654bd4554bf9    250MB    80.4MB
hyperledger/fabric-baseos:latest            654bd4554bf9    250MB    80.4MB
hyperledger/fabric-ca:1.5                  453a0a727e82    381MB    123MB
hyperledger/fabric-ca:1.5.17                453a0a727e82    381MB    123MB
hyperledger/fabric-ca:latest                453a0a727e82    381MB    123MB
hyperledger/fabric-ccenv:2.5                86dded7eda92    1GB      255MB
hyperledger/fabric-ccenv:2.5.15             86dded7eda92    1GB      255MB
hyperledger/fabric-ccenv:latest              86dded7eda92    1GB      255MB
hyperledger/fabric-orderer:2.5              9972c209be47    178MB    49.5MB
hyperledger/fabric-orderer:2.5.15           9972c209be47    178MB    49.5MB
hyperledger/fabric-orderer:latest            9972c209be47    178MB    49.5MB
hyperledger/fabric-peer:2.5                 3e25adc31a75    230MB    66.2MB
hyperledger/fabric-peer:2.5.15              3e25adc31a75    230MB    66.2MB
hyperledger/fabric-peer:latest               3e25adc31a75    230MB    66.2MB
g2022_anuprita_mhapankar@cloudshell:~/fabric-samples$
```


Step 6: Verify Docker images

docker images | grep fabric

```
g2022_anuprita_mhapankar@cloudshell:~/fabric-samples$ docker images | grep fabric
WARNING: This output is designed for human readability. For machine-readable output, please use --format.
ghcr.io/hyperledger/fabric-baseos:2.5.15      654bd4554bf9      250MB      80.4MB
ghcr.io/hyperledger/fabric-ca:1.5.17          453a0a727e82      381MB      123MB
ghcr.io/hyperledger/fabric-ccenv:2.5.15       86dded7eda92      1GB        255MB
ghcr.io/hyperledger/fabric-orderer:2.5.15     9972c209be47      178MB      49.5MB
ghcr.io/hyperledger/fabric-peer:2.5.15        3e25adc31a75      230MB      66.2MB
hyperledger/fabric-baseos:2.5                 654bd4554bf9      250MB      80.4MB
hyperledger/fabric-baseos:2.5.15              654bd4554bf9      250MB      80.4MB
hyperledger/fabric-baseos:latest              654bd4554bf9      250MB      80.4MB
hyperledger/fabric-ca:1.5                     453a0a727e82      381MB      123MB
hyperledger/fabric-ca:1.5.17                  453a0a727e82      381MB      123MB
hyperledger/fabric-ca:latest                   453a0a727e82      381MB      123MB
hyperledger/fabric-ccenv:2.5                   86dded7eda92      1GB        255MB
hyperledger/fabric-ccenv:2.5.15                86dded7eda92      1GB        255MB
hyperledger/fabric-ccenv:latest                86dded7eda92      1GB        255MB
hyperledger/fabric-orderer:2.5                 9972c209be47      178MB      49.5MB
hyperledger/fabric-orderer:2.5.15              9972c209be47      178MB      49.5MB
hyperledger/fabric-orderer:latest              9972c209be47      178MB      49.5MB
hyperledger/fabric-peer:2.5                    3e25adc31a75      230MB      66.2MB
hyperledger/fabric-peer:2.5.15                 3e25adc31a75      230MB      66.2MB
hyperledger/fabric-peer:latest                 3e25adc31a75      230MB      66.2MB
g2022_anuprita_mhapankar@cloudshell:~/fabric-samples$
```

Step 7: Set Environment Variables

```
export PATH=$PATH:$HOME/fabric-samples/bin
```

```
export FABRIC_CFG_PATH=$HOME/fabric-samples/config
```

```
peer version
```

```
g2022_anuprita_mhapankar@cloudshell:~/fabric-samples$ export PATH=$PATH:$HOME/fabric-samples/bin
g2022_anuprita_mhapankar@cloudshell:~/fabric-samples$ export FABRIC_CFG_PATH=$HOME/fabric-samples/config
g2022_anuprita_mhapankar@cloudshell:~/fabric-samples$ peer version
peer:
  Version: v2.5.15
  Commit SHA: 83c7930
  Go version: go1.26.0
  OS/Arch: linux/amd64
  Chaincode:
    Base Docker Label: org.hyperledger.fabric
    Docker Namespace: hyperledger

g2022_anuprita_mhapankar@cloudshell:~/fabric-samples$
```

Step 8: Start the Network and Create Channel

```
cd $HOME/fabric-samples/test-network
```

```
./network.sh up createChannel
```

```

/home/g2022_anuprita_mhapankar/fabric-samples/test-network/./bin/configtxgen
+ '[' 0 -eq 1 ']'
+ configtxgen -profile ChannelUsingRaft -outputBlock ./channel-artifacts/mychannel.block -channelID mychannel
2026-04-09 14:56:55.392 UTC 0001 INFO [common.tools.configtxgen] main -> Loading configuration
2026-04-09 14:56:55.402 UTC 0002 INFO [common.tools.configtxgen.localconfig] completeInitialization -> orderer type: etcdraft
2026-04-09 14:56:55.402 UTC 0003 INFO [common.tools.configtxgen.localconfig] completeInitialization -> Orderer.EtcdRaft.Options unset, sett
n_tick:10 heartbeat_tick:1 max_inflight_blocks:5 snapshot_interval_size:16777216
2026-04-09 14:56:55.402 UTC 0004 INFO [common.tools.configtxgen.localconfig] Load -> Loaded configuration: /home/g2022_anuprita_mhapankar/f
configtx.yaml
2026-04-09 14:56:55.408 UTC 0005 INFO [common.tools.configtxgen] doOutputBlock -> Generating genesis block
2026-04-09 14:56:55.408 UTC 0006 INFO [common.tools.configtxgen] doOutputBlock -> Creating application channel genesis block
2026-04-09 14:56:55.410 UTC 0007 INFO [common.tools.configtxgen] doOutputBlock -> Writing genesis block
+ res=0
Creating channel mychannel
Adding orderers
+ . scripts/orderer.sh mychannel
+ '[' 0 -eq 1 ']'
+ res=0
Status: 201
{
    "name": "mychannel",
    "url": "/participation/v1/channels/mychannel",
    "consensusRelation": "consenter",
    "status": "active",
    "height": 1
}

Channel 'mychannel' created
Joining org1 peer to the channel...
Using organization 1
+ peer channel join -b ./channel-artifacts/mychannel.block
+ res=0

```

```
g2022_anuprita_mhapankar@cloudshell:~/fabric-samples/test-network$
```

Step 9: Deploy Chaincode

```
./network.sh deployCC \  
-ccn basic \  
-ccp ../asset-transfer-basic/chaincode-javascript \  
-ccl javascript
```

```
g2022_anuprita_mhapankar@cloudshell:~/fabric-samples/test-network$ cd $HOME/fabric-samples/test-network  
  
./network.sh deployCC \  
-ccn basic \  
-ccp ../asset-transfer-basic/chaincode-javascript \  
-ccl javascript  
Using docker and docker-compose  
deploying chaincode on channel 'mychannel'  
executing with the following  
- CHANNEL_NAME: mychannel  
- CC_NAME: basic  
- CC_SRC_PATH: ../asset-transfer-basic/chaincode-javascript  
- CC_SRC_LANGUAGE: javascript  
- CC_VERSION: 1.0  
- CC_SEQUENCE: auto  
- CC_END_POLICY: NA  
- CC_COLL_CONFIG: NA  
- CC_INIT_FCN: NA  
- DELAY: 3  
- MAX_RETRY: 5  
- VERBOSE: false  
executing with the following  
- CC_NAME: basic  
- CC_SRC_PATH: ../asset-transfer-basic/chaincode-javascript  
- CC_SRC_LANGUAGE: javascript  
- CC_VERSION: 1.0  
+ '[' false = true ']'  
+ peer lifecycle chaincode package basic.tar.gz --path ../asset-transfer-basic/chaincode-javascript --lang node --label basic_1.0  
+ res=0  
Chaincode is packaged  
Installing chaincode on peer0.org1...  
Using organization 1
```

```
Checking the commit readiness of the chaincode definition successful on peer0.org2 on channel 'mychannel'  
Using organization 1  
Using organization 2  
+ peer lifecycle chaincode commit -o localhost:7050 --ordererTLSHostnameOverride orderer.example.com --tls --cafile /home/g2022_anuprita_mhapankar/fabric-samples/test-network/ordererOrganizations/example.com/tlsca/tlsca.example.com-cert.pem --channelID mychannel --name basic --peerAddresses localhost:7051 --tlsRootCertFiles /home/g2022_anuprita_mhapankar/fabric-samples/test-network/organizations/peerOrganizations/org1.example.com/tlsca/tlsca.org1.example.com-cert.pem --peerAddresses localhost:9051 --tlsRootCertFiles /home/g2022_anuprita_mhapankar/fabric-samples/test-network/organizations/peerOrganizations/org2.example.com/tlsca/tlsca.org2.example.com-cert.pem --version 1  
sequence 1  
+ res=0  
2026-04-09 15:02:00.641 UTC 0001 INFO [chaincodeCmd] ClientWait -> txid [d578fad7d5ea9fbbab1c00b0f3bb78e4c57add2a11c8387b9bed0a90efd9559c] committed with status (VALID)  
alhost:7051  
2026-04-09 15:02:00.646 UTC 0002 INFO [chaincodeCmd] ClientWait -> txid [d578fad7d5ea9fbbab1c00b0f3bb78e4c57add2a11c8387b9bed0a90efd9559c] committed with status (VALID)  
alhost:9051  
Chaincode definition committed on channel 'mychannel'  
Using organization 1  
Querying chaincode definition on peer0.org1 on channel 'mychannel'...  
Attempting to query committed status on peer0.org1, Retry after 3 seconds.  
+ peer lifecycle chaincode querycommitted --channelID mychannel --name basic  
+ res=0  
Committed chaincode definition for chaincode 'basic' on channel 'mychannel':  
Version: 1.0, Sequence: 1, Endorsement Plugin: escc, Validation Plugin: vscc, Approvals: [Org1MSP: true, Org2MSP: true]  
Query chaincode definition successful on peer0.org1 on channel 'mychannel'  
Using organization 2  
Querying chaincode definition on peer0.org2 on channel 'mychannel'...  
Attempting to query committed status on peer0.org2, Retry after 3 seconds.  
+ peer lifecycle chaincode querycommitted --channelID mychannel --name basic  
+ res=0  
Committed chaincode definition for chaincode 'basic' on channel 'mychannel':  
Version: 1.0, Sequence: 1, Endorsement Plugin: escc, Validation Plugin: vscc, Approvals: [Org1MSP: true, Org2MSP: true]  
Query chaincode definition successful on peer0.org2 on channel 'mychannel'  
Chaincode initialization is not required  
g2022_anuprita_mhapankar@cloudshell:~/fabric-samples/test-network$
```

Step 10: Set Peer Environment Variables

```
export FABRIC_CFG_PATH=$HOME/fabric-samples/config  
export PATH=$PATH:$HOME/fabric-samples/bin  
export CORE_PEER_TLS_ENABLED=true  
export CORE_PEER_LOCALMSPID="Org1MSP"  
  
export  
CORE_PEER_TLS_ROOTCERT_FILE=$HOME/fabric-samples/test-network/organization  
s/peerOrganizations/org1.example.com/peers/peer0.org1.example.com/tls/ca.c  
rt
```



```
export
CORE_PEER_MSPCONFIGPATH=$HOME/fabric-samples/test-network/organizations/peerOrganizations/org1.example.com/users/Admin@org1.example.com/msp
export CORE_PEER_ADDRESS=localhost:7051
```

```
g2022_anuprita_mhapankar@cloudshell:~/fabric-samples/test-network$ export FABRIC_CFG_PATH=$HOME/fabric-samples/config
export PATH=$PATH:$HOME/fabric-samples/bin
export CORE_PEER_TLS_ENABLED=true
export CORE_PEER_LOCALMSPID="Org1MSP"
export CORE_PEER_TLS_ROOTCERT_FILE=$HOME/fabric-samples/test-network/organizations/peerOrganizations/org1.example.com/peers/peer0.org1.example.com/tls/ca.crt
export CORE_PEER_MSPCONFIGPATH=$HOME/fabric-samples/test-network/organizations/peerOrganizations/org1.example.com/users/Admin@org1.example.com/msp
export CORE_PEER_ADDRESS=localhost:7051
g2022_anuprita_mhapankar@cloudshell:~/fabric-samples/test-network$
```

Step 11: Initialize Ledger

```
peer chaincode invoke \
  -o localhost:7050 \
  --ordererTLSHostnameOverride orderer.example.com \
  --tls \
  --cafile
$HOME/fabric-samples/test-network/organizations/ordererOrganizations/example.com/orderers/orderer.example.com/msp/tlscacerts/tlsca.example.com-cert.pem \
  -C mychannel \
  -n basic \
  --peerAddresses localhost:7051 \
  --tlsRootCertFiles
$HOME/fabric-samples/test-network/organizations/peerOrganizations/org1.example.com/peers/peer0.org1.example.com/tls/ca.crt \
  --peerAddresses localhost:9051 \
  --tlsRootCertFiles
$HOME/fabric-samples/test-network/organizations/peerOrganizations/org2.example.com/peers/peer0.org2.example.com/tls/ca.crt \
  -c '{"function":"InitLedger","Args":[]}'
```

```
g2022_anuprita_mhapankar@cloudshell:~/fabric-samples/test-network$ peer chaincode invoke \
-o localhost:7050 \
--ordererTLSHostnameOverride orderer.example.com \
--tls \
--cafile $HOME/fabric-samples/test-network/organizations/ordererOrganizations/example.com/orderers/orderer.example.com/msp/tlscacerts/tlsca.example.com-cert.pem \
-C mychannel \
-n basic \
--peerAddresses localhost:7051 \
--tlsRootCertFiles $HOME/fabric-samples/test-network/organizations/peerOrganizations/org1.example.com/peers/peer0.org1.example.com/tls/ca.crt \
--peerAddresses localhost:9051 \
--tlsRootCertFiles $HOME/fabric-samples/test-network/organizations/peerOrganizations/org2.example.com/peers/peer0.org2.example.com/tls/ca.crt \
-c '{"function":"InitLedger","Args":[]}'
g2022_04-06 13:04:00.450 UTC INFO [chaincodeCmd] chaincodeInvokeOrQuery -> Chaincode invoke successful. result: status:200
g2022_anuprita_mhapankar@cloudshell:~/fabric-samples/test-network$
```

Step 12: Query All Assets

```
peer chaincode query -C mychannel -n basic -c '{"Args":["GetAllAssets"]}'
```

```
g2022_anuprita_mhapankar@cloudshell:~/fabric-samples/test-network$ peer chaincode query -C mychannel -n basic -c '{"Args":["GetAllAssets"]}'
[{"AppraisedValue":300,"Color":"blue","ID":"asset1","Owner":"Tomoko","Size":5,"docType":"asset"}, {"AppraisedValue":400,"Color":"red","ID":"asset2","Owner":"Brad","Size":5,"docType":"asset"}, {"AppraisedValue":500,"Color":"green","ID":"asset3","Owner":"Jin Soo","Size":10,"docType":"asset"}, {"AppraisedValue":600,"Color":"yellow","ID":"asset4","Owner":"Max","Size":10,"docType":"asset"}, {"AppraisedValue":700,"Color":"black","ID":"asset5","Owner":"Adriana","Size":15,"docType":"asset"}, {"AppraisedValue":800,"Color":"white","ID":"asset6","Owner":"Michel","Size":15,"docType":"asset"}]
g2022_anuprita_mhapankar@cloudshell:~/fabric-samples/test-network$
```

Step 13: Read a Specific Asset

```
peer chaincode query -C mychannel -n basic -c
 '{"Args":["ReadAsset","asset5"]}'
```

```
g2022_anuprita_mhapankar@cloudshell:~/fabric-samples/test-network$ peer chaincode query -C mychannel -n basic -c '{"Args":["ReadAsset","asset5"]}'
{"AppraisedValue":700,"Color":"black","ID":"asset5","Owner":"Adriana","Size":15,"docType":"asset"}
g2022_anuprita_mhapankar@cloudshell:~/fabric-samples/test-network$
```


Step 14: Create a New Asset

```
peer chaincode invoke \  
  -o localhost:7050 \  
  --ordererTLSHostnameOverride orderer.example.com \  
  --tls \  
  --cafile  
$HOME/fabric-samples/test-network/organizations/ordererOrganizations/example.com/orderers/orderer.example.com/msp/tlscacerts/tlsca.example.com-cert.pem \  
  -C mychannel \  
  -n basic \  
  --peerAddresses localhost:7051 \  
  --tlsRootCertFiles  
$HOME/fabric-samples/test-network/organizations/peerOrganizations/org1.example.com/peers/peer0.org1.example.com/tls/ca.crt \  
  --peerAddresses localhost:9051 \  
  --tlsRootCertFiles  
$HOME/fabric-samples/test-network/organizations/peerOrganizations/org2.example.com/peers/peer0.org2.example.com/tls/ca.crt \  
  -C  
'{"function":"CreateAsset","Args":["asset7","blue","10","Anuprita","500"]}'  
,
```

```
q2022_anuprita_mhapankar@cloudshell:~/fabric-samples/test-network$ peer chaincode invoke \  
-o localhost:7050 \  
--ordererTLSHostnameOverride orderer.example.com \  
--tls \  
--cafile $HOME/fabric-samples/test-network/organizations/ordererOrganizations/example.com/orderers/orderer.example.com/msp/tlscacerts/tlsca.example.com-cert.pem \  
-C mychannel \  
-n basic \  
--peerAddresses localhost:7051 \  
--tlsRootCertFiles $HOME/fabric-samples/test-network/organizations/peerOrganizations/org1.example.com/peers/peer0.org1.example.com/tls/ca.crt \  
--peerAddresses localhost:9051 \  
--tlsRootCertFiles $HOME/fabric-samples/test-network/organizations/peerOrganizations/org2.example.com/peers/peer0.org2.example.com/tls/ca.crt \  
-C '{"function":"CreateAsset","Args":["asset7","blue","10","Anuprita","500"]}'  
$ peer chaincode invoke -o localhost:7050 --ordererTLSHostnameOverride orderer.example.com --tls --cafile $HOME/fabric-samples/test-network/organizations/ordererOrganizations/example.com/orderers/orderer.example.com/msp/tlscacerts/tlsca.example.com-cert.pem -C mychannel -n basic --peerAddresses localhost:7051 --tlsRootCertFiles $HOME/fabric-samples/test-network/organizations/peerOrganizations/org1.example.com/peers/peer0.org1.example.com/tls/ca.crt --peerAddresses localhost:9051 --tlsRootCertFiles $HOME/fabric-samples/test-network/organizations/peerOrganizations/org2.example.com/peers/peer0.org2.example.com/tls/ca.crt -C '{"function":"CreateAsset","Args":["asset7","blue","10","Anuprita","500"]}'  
Chaincode invoke successful. result: status:200 payload:{"ID":"asset7","Color":"blue","Size":10,"Owner":"Anuprita","AppraisedValue":500}  
q2022_anuprita_mhapankar@cloudshell:~/fabric-samples/test-network$
```

Step 15: Transfer Asset

```
peer chaincode invoke \  
  -o localhost:7050 \  
  --ordererTLSHostnameOverride orderer.example.com \  
  --tls \  
  --cafile  
$HOME/fabric-samples/test-network/organizations/ordererOrganizations/example.com/orderers/orderer.example.com/msp/tlscacerts/tlsca.example.com-cert.pem \  
  -C mychannel \  
  -n basic \  
  --peerAddresses localhost:7051 \  
  --tlsRootCertFiles  
$HOME/fabric-samples/test-network/organizations/peerOrganizations/org1.example.com/peers/peer0.org1.example.com/tls/ca.crt \  
  --peerAddresses localhost:9051 \  
  --tlsRootCertFiles  
$HOME/fabric-samples/test-network/organizations/peerOrganizations/org2.example.com/peers/peer0.org2.example.com/tls/ca.crt
```

```
--peerAddresses localhost:9051 \
--tlsRootCertFiles
$HOME/fabric-samples/test-network/organizations/peerOrganizations/org2.example.com/peers/peer0.org2.example.com/tls/ca.crt \
-c '{"function": "TransferAsset", "Args": ["asset7", "Sneha"]}'
```

```
g2022_anuprita_mhapankar@cloudshell:~/fabric-samples/test-network$ peer chaincode invoke \
--no localhost:7050 \
--ordererTLSHostnameOverride orderer.example.com \
--tls \
--cafile $HOME/fabric-samples/test-network/organizations/ordererOrganizations/example.com/orderers/orderer.example.com/msp/tlscacerts/tlsca.example.com-cert.pem \
-C mychannel \
-n basic \
--peerAddresses localhost:7051 \
--tlsRootCertFiles $HOME/fabric-samples/test-network/organizations/peerOrganizations/org1.example.com/peers/peer0.org1.example.com/tls/ca.crt \
--peerAddresses localhost:9051 \
--tlsRootCertFiles $HOME/fabric-samples/test-network/organizations/peerOrganizations/org2.example.com/peers/peer0.org2.example.com/tls/ca.crt \
-c '{"function": "TransferAsset", "Args": ["asset7", "Sneha"]}'
2022-04-05 15:04:55.967 UTC [INFO] [chaincodeCmd] chaincodeInvokeOrQuery -> Chaincode invoke successful. result: status:200 payload:"Anuprita"
g2022_anuprita_mhapankar@cloudshell:~/fabric-samples/test-network$
```

Step 16: Stop the Network

```
cd $HOME/fabric-samples/test-network
./network.sh down
```

```
g2022_anuprita_mhapankar@cloudshell:~/fabric-samples/test-network$ cd $HOME/fabric-samples/test-network
./network.sh down
Using docker and docker-compose
Stopping network
Project name can't be empty. Use '--project-name' to set a valid name
Error response from daemon: get docker_orderer.example.com: no such volume
Error response from daemon: get docker_peer0.org1.example.com: no such volume
Error response from daemon: get docker_peer0.org2.example.com: no such volume
Removing remaining containers
c72f6a9969f0
a811e27601bb
5f5198e28eca
3bcfb908afe1
8d2fc8e74375
Removing deprecated chaincode docker images
Untagged: dev-peer0.org2.example.com-basic_1.0-1d4d445573112813889f87c307bc2b5fbc664c87b24c035beel5cbcc7f59b629-c68f8ecc2a0ff0cc1497615f5e33a8bfa908e0fd3555f864cf1398b52d341c
b:latest
Deleted: sha256:fa120af33c4db851b5dded871dd45c39e7e3567eb3d753fccfca850a32284087
Untagged: dev-peer0.org1.example.com-basic_1.0-1d4d445573112813889f87c307bc2b5fbc664c87b24c035beel5cbcc7f59b629-d3cf9b1f4b9747cc8433ac0e74bfaeade1cf0b97f5c1163835f604aa885d60d
c:latest
Deleted: sha256:edc7441a9a8c48b025dba4a79878631a900a26be317335ae9c9b2512c869f810
g2022_anuprita_mhapankar@cloudshell:~/fabric-samples/test-network$
```

CONCLUSION:

The Hyperledger Fabric network was successfully implemented and chaincode was deployed. Asset operations such as creation, querying, and transfer were performed successfully. This experiment demonstrated secure and efficient asset management using a permissioned blockchain system.